Dylan Vazquez

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EDUCATION

Inter American University of Puerto Rico — Bayamon Campus *Aug 2024 – May 2028*
B.S. in Computer Engineering · Minor in Systems Engineering Control, Robotics & Automation
GPA: 3.20 · Expected Graduation: May 2028
Relevant Coursework: Software Design & Construction, Introduction to Programming, Advanced Programming, Digital System Design, Differential Equations, Probability & Statistics, Cybersecurity

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, SQL, Verilog
Machine Learning & Data Analysis: scikit-learn, NumPy, pandas, Matplotlib, NetworkX, regression, clustering, feature engineering, predictive modeling
Development & Version Control: Git, GitHub, GitLab, VSCodium, Streamlit, JavaFX, FXML, Scene Builder
Digital Systems & Simulation: Xilinx Vivado, Verilog-based FPGA design and simulation, NI Multisim, Logisim
Cybersecurity & Digital Forensics: Volatility 3, OpenStego, Wireshark, Nmap, Scapy, hping3, Kali Linux, VirtualBox, memory forensics, steganography, packet analysis, network traffic analysis
Mathematics & Engineering: Differential equations, numerical analysis, circuit simulation, Desmos
Productivity & Design: Microsoft Excel, Word, PowerPoint, Notion, Figma, Canva
Languages: Spanish, English

ENGINEERING & LEADERSHIP EXPERIENCE

LLM & NLP Research Team Member — Raytheon Autonomous Vehicle Competition *Sep 2026 – Present*
- Researching LLM-based natural language interfaces for interpreting written instructions and transcribed speech.
- Investigating operator intent extraction and translation of natural language into structured vehicle commands.
- Collaborating on an autonomous vehicle research effort integrating NLP with unmanned system control.

CAHSI Southeast Advocate — Computing Alliance of Hispanic-Serving Institutions *Aug 2026 – Present*
- Serve as a student connector, promoting computing opportunities, scholarships, internships, and professional development.
- Support student engagement, technical workshops, and collaboration with campus organizations.
- Participate in regional advocate coordination and outreach initiatives to strengthen the computing community.

SCIICOM Vocal — Student Chapter of the Institute of Computer Engineers *Sep 2026 – Present*
- Support student chapter activities, technical initiatives, and professional development opportunities.
- Contribute to organizational planning, communication, and student engagement.

HPE CDS Tech Challenge — Team Member *Sep 2026 – Present*
- Participate in a team-based technology competition addressing real-world computing and engineering challenges.
- Collaborate on technical preparation, problem-solving, and solution development for upcoming challenges.

Big Brothers — Student Mentoring Projects (Volunteer) *Aug 2023 – May 2024*
- Led interactive activities encouraging student engagement and teamwork.
- Modeled respectful communication and designed engaging student activities.

TECHNICAL & ACADEMIC PROJECTS

Puerto Rico Karst & Caves GIS Portfolio Project — Collaborative Geospatial Analysis
- Collaborating on a QGIS-based study of Puerto Rico's north-coast karst, sinkholes, cave entrances, hydrography, geology, and aquifers using public USGS, NOAA, and Puerto Rico datasets.

CPU Thermal Modeling & Machine Learning — Newton's Law of Cooling
Python · Differential Equations · Regression · Random Forest
- Analyzed 7,000+ CPU thermal records, engineering time-series features from temperature, power, frequency, and fan speed.
- Compared physics-based cooling models with Linear Regression and Random Forest.
- Achieved lowest prediction error with Linear Regression (MAE 0.040; RMSE 0.050).

Machine Learning — NBA Draft Prediction
Python · scikit-learn · NetworkX
- Collaborated on an ML model predicting the likelihood of NBA prospects becoming Top 3 draft picks.
- Developed data cleaning, feature engineering, graph-based similarity, and regression pipelines.
- Evaluated models using train-test splits, correlation analysis, and prediction visualizations.

Firmware Corruption & Network Attack Simulation — Cybersecurity Testing & Analysis
Kali Linux · Ubuntu · VirtualBox · Wireshark · Scapy · hping3 · TFTP
- Authored a cybersecurity lab guide simulating firmware transfer disruption in an isolated virtual network.
- Designed TFTP transfer, packet injection, and denial-of-service tests aligned with MITRE ATT&CK T1495.
- Incorporated Wireshark captures, integrity checks, and traffic comparisons to evaluate attack effects.

AI-Assisted Oscilloscope & Intelligent Signal Analyzer — AI Agent Integration

Generative AI · LLM Integration · Prompt Engineering · Signal Interpretation

- Developing an AI interpretation agent for an Arduino UNO R4-based educational oscilloscope.
- Integrating measured waveform characteristics into prompts for natural-language signal explanations.
- Coordinating AI integration with the circuit's signal-acquisition and analysis workflow.

Topic-Based Bible Verse Retrieval Application

Python · Text Processing · File Handling

- Built a Python application searching 66 Bible books using keyword associations across 15 topics.
- Implemented whole-word matching, randomized verse selection, and interactive topic navigation.

Java Tic-Tac-Toe — Collaborative Project

Java · JavaFX · FXML · Scene Builder

- Collaborated on a graphical Tic-Tac-Toe game with turn sequencing, win detection, and event-driven controls.
- Applied MVC architecture to separate game logic, JavaFX controllers, and FXML interface components.

Dungeon RPG

C++ · Object-Oriented Programming · STL

- Developed a 1,000+ line console RPG with object-oriented combat, encounters, and character progression.
- Implemented inventory management and file-based save/load using STL vectors and maps.